

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a029a5f27357d3dea.jsonl`  
- SHA-256: `0c52f699158562e05bc377bb301597d37ad9db2facd42d64878c6d92399cc7f6`  
- Source modified: `2026-05-31T04:22:16+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.033Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On the olmOCR-Bench evaluation, olmOCR 2 scores 84.9 on tables and 83.7 on multi-column layouts, with the paper attributing its largest gains specifically to table parsing and multi-column layouts ? the exact document features muneem needs for dense, multi-column bank statements."

****Source:**** <https://arxiv.org/html/2510.19817v1> (primary)

****Supporting quote:**** "largest improvements in math formula conversion, table parsing, and multi-column layouts"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.182Z)

I'll verify this claim by examining the primary source (the olmOCR 2 paper) and checking the specific numbers cited.

3. assistant (2026-05-31T04:21:13.715Z)

I'll fetch the primary source to verify the specific numbers claimed: 84.9 on tables and 83.7 on multi-column, plus the attribution quote.

4. user (2026-05-31T04:21:21.112Z)

olmOCR 2 Benchmark Results

Key Scores from Table 3

For **olmOCR 2** (final row in the results table):

- **Tables category**: 84.9
- **Multi-column category**: 83.7
- **Overall score**: 82.4 ± 1.1

Per-Category Scores (Complete)

The full olmOCR 2 results across all categories in Table 3:

Category	Score
ArXiv	83.0
Old scans math	82.3
Tables	84.9
Old scans	47.7
Headers & footers	96.1
Multi column	83.7
Long tiny text	81.9
Base	99.7
Overall	82.4 ± 1.1

Statement on Largest Improvements

The abstract states: "the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

5. user (2026-05-31T04:21:23.097Z)

Web search results for query: "olmOCR 2 olmOCR-Bench tables 84.9 multi-column 83.7 scores"

Links: [{"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "(PDF) olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"}, {"title": "olmOCR 2 Unit Test Rewards for Document OCR", "url": "https://arxiv.org/html/2510.19817v1"}, {"title": "DeepSeek-OCR-OmniDocBench/olmocr/README.md at main · alphaXiv/DeepSeek-OCR-OmniDocBench", "url": "https://github.com/alphaXiv/DeepSeek-OCR-OmniDocBench/blob/main/olmocr/README.md"}, {"title": "GitHub - allenai/olmocr: Toolkit for linearizing PDFs for LLM datasets/training · GitHub", "url": "https://github.com/allenai/olmocr"}, {"title": "[2510.19817] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://arxiv.org/abs/2510.19817"}, {"title": "olmocr/olmocr/bench at main · allenai/olmocr", "url": "https://github.com/allenai/olmocr/tree/main/olmocr/bench"}]

Based on the search results, I found information about olmOCR 2 and its performance scores on the olmOCR-Bench evaluation:

olmOCR 2 achieves 84.9% on tables (up from 72.9%) and 83.7% on multi-column (up from 77.3%) on olmOCR-Bench. Overall, olmOCR 2 achieves 82.4 points on olmOCR-Bench?a nearly +4-point improvement over the previous release.

olmOCR 2 is powered by olmOCR-2-7B-1025, a specialized 7B vision language model trained using reinforcement learning with verifiable rewards (RLVR), where rewards are a diverse set of binary unit tests. RL training on test cases results in state-of-the-art performance on olmOCR-Bench, with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:25.975Z)

The numbers are confirmed by the primary arXiv source. Let me cross-check the attribution quote and the "largest improvements" wording with the official Ai2 blog to verify the claim isn't an overreach.

7. user (2026-05-31T04:21:31.981Z)

olmOCR 2 Performance Improvements

Largest gains:

"olmOCR 2 delivers substantial gains where OCR traditionally struggles: 82.3% on old math scans (up from 79.9%)"

Table performance:

"84.9% on tables (up from 72.9%)"

Multi-column performance:

"83.7% on multi-column (up from 77.3%)"

The document indicates that table recognition and multi-column layout handling saw the most dramatic improvements? a 12-point gain for tables and a 6.4-point improvement for multi-column documents.

8. assistant (2026-05-31T04:21:44.084Z)

I now have enough to evaluate the claim rigorously. Let me assess each component.

Numbers verification:

- Tables = 84.9 ? CONFIRMED by primary arXiv source (Table 3) and Ai2 blog.
- Multi-column = 83.7 ? CONFIRMED by both.
- These are olmOCR-Bench scores ? CONFIRMED.

Attribution quote verification:

- The claimed quote "largest improvements in math formula conversion, table parsing, and multi-column layouts" is verbatim accurate from the abstract.

Now the skeptical scrutiny ? is the claim an overreach?

The claim asserts the paper attributes its "largest gains specifically to table parsing and multi-column layouts." Let me check the precision here:

1. The actual paper wording is "the largest improvements in **math formula conversion**, **table parsing**, **and multi-column layouts**." The claim drops "math formula conversion" and frames table+multi-column as the headline. This is a minor selective emphasis, but the claim does say "largest gains specifically to table parsing and multi-column layouts" ? which omits that math formula conversion is co-equally credited. This is a slight cherry-pick but not a misrepresentation: table parsing and multi-column ARE explicitly named among the largest gains.

2. Magnitude check: Tables went 72.9 ? 84.9 (+12.0), multi-column 77.3 ? 83.7 (+6.4). The blog itself says "table recognition and multi-column layout handling saw the most dramatic improvements? a 12-point gain for tables." So the +12 table gain genuinely IS the largest single-category jump. This supports the claim's emphasis.

3. The claim's relevance framing ("the exact document features muneem needs") is the researcher's own editorial connection, not a paper claim ? that's fine, it's contextual.

The one genuine caveat for skepticism: The "Old scans" score is 47.7 ? very low. And these are benchmark numbers on olmOCR-Bench (Ai2's own benchmark), which is a self-reported benchmark from the model authors. But the claim only asserts what the paper says about its own benchmark, and accurately reports those numbers. The claim is appropriately scoped ("On the olmOCR-Bench

evaluation...").

Let me do one contradiction check to ensure no independent source disputes these figures.

9. user (2026-05-31T04:21:53.734Z)

Web search results for query: "olmOCR 2 independent evaluation tables hallucination accuracy comparison OmniDocBench"

Links: [{"title": "olmOCR 2 Unit Test Rewards for Document OCR", "url": "https://arxiv.org/html/2510.19817v1"}, {"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "(PDF) olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"}, {"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "olmOCR: Unlocking Trillions of Tokens in PDFs with Vision Language Models", "url": "https://olmocr.allenai.org/papers/olmocr.pdf"}, {"title": "OmniDocBench is Saturated, What's Next for OCR Benchmarks?", "url": "https://www.llamaindex.ai/blog/omnidocbench-is-saturated-what-s-next-for-ocr-benchmarks"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "Papers Explained 480: olmOCR 2 | by Ritvik Rastogi | Medium", "url": "https://ritvik19.medium.com/papers-explained-480-olmocr-2-33fbf89757eb"}]

Based on the search results, I found relevant information about olmOCR 2 and its evaluation:

olmOCR 2 Performance and Evaluation

olmOCR 2 achieves 82.4 points on olmOCR-Bench, representing a nearly 4-point improvement over the previous release. olmOCR 2 outperforms specialized tools like Marker (76.1) and MinerU (75.8).

Hallucination Robustness

Regarding hallucinations, the pipeline is robust to hallucinations, as even if Claude makes an error during OCR, that does not affect the pipeline since the HTML output alone is used to generate unit tests. Additionally, residual instances of repeated n-grams or hallucinated text in long-tiny-text settings have been identified as common failure modes.

Evaluation Framework

olmOCR-Bench measures performance through unit test cases for each document, including text presence/absence checks, natural reading order verification, table accuracy checks, math formula accuracy, and baseline robustness checks.

Comparison with OmniDocBench

While both benchmarks are used for evaluating OCR systems, OmniDocBench uses Normalized Edit Distance for text, Tree-Edit-Distance-based similarity (TEDS) for tables, and Character Detection Matching (CDM) for formulas, which differs from olmOCR-Bench's unit test approach.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-05-31T04:22:04.575Z)

I have sufficient evidence to render a verdict.

****Assessment:****

The factual core of the claim is fully verified against the primary source:

- Tables = 84.9 ? (arXiv Table 3 + Ai2 blog confirm)

- Multi-column = 83.7 ? (both sources confirm)
- These are olmOCR-Bench scores ?
- The attribution quote "largest improvements in math formula conversion, table parsing, and multi-column layouts" ? (verbatim from abstract)
- Magnitude supports emphasis: tables +12.0 (72.9?84.9) is the single largest category gain; multi-column +6.4. The Ai2 blog independently states these "saw the most dramatic improvements."

****Skeptical scrutiny applied:****

1. ***Selective quoting?*** The claim drops "math formula conversion" from the trio and headlines table+multi-column. This is mild selective emphasis, but NOT a misrepresentation ? both features the claim names are explicitly among the paper's listed largest gains, and tables is genuinely the largest single jump (+12). The framing toward muneem's needs is legitimate contextual relevance, not a fabricated paper claim.

2. ***Source quality:*** Primary arXiv source + corroborating official blog. Matches claim strength.

3. ***Self-reported benchmark caveat:*** These are scores on olmOCR-Bench, Ai2's own benchmark ? a real limitation for the broader research, BUT the claim is correctly scoped ("On the olmOCR-Bench evaluation..."). It does not overclaim independent validation. No contradicting source disputes the numbers.

4. ***Currency:*** Paper is Oct 2025 (arXiv 2510.19817), current as of May 2026.

The claim is accurate, properly scoped, primary-sourced, and current. The only blemish is mild emphasis-shifting (omitting math from the "largest gains" list), which is not enough to refute ? the named features are correctly attributed and the table gain is factually the largest.

11. user (2026-05-31T04:22:11.698Z)

Structured output provided successfully

12. assistant (2026-05-31T04:22:16.430Z)

I've completed the verification. The claim is ****NOT refuted**** (refuted=false) with high confidence.

The factual core ? Tables=84.9, Multi-column=83.7 on olmOCR-Bench, plus the verbatim "largest improvements in math formula conversion, table parsing, and multi-column layouts" quote ? is fully confirmed by both the primary arXiv source (2510.19817v1, Table 3) and the official Ai2 blog. The +12.0 table gain is genuinely the largest single-category jump, supporting the claim's emphasis.

The only blemishes I found (mild emphasis-shifting that drops "math formula conversion" from the paper's three-item list, and the fact that olmOCR-Bench is the authors' own benchmark) are not strong enough to refute, since the claim correctly names two of the three credited features and is properly scoped to "the olmOCR-Bench evaluation."